Wenhui Yan | 闫文慧

wenhuiyan@cau.edu.cn | wenhui-yan.github.io | Wenhui-Yan | Google Scholar | ResearchGate

Education

China Agricultural University (985), Beijing, China — **Supervisor: Prof. Yelu Zeng**

M.S. Land Use and Information Technology (3.55/4.0) — Aug 2025 – Jun 2027 (Expected)

Core Courses: The Theory and Method on GIS (95), Artificial Intelligence and Geoscience Applications (94), Quantitative Remote Sensing Technology and Application (92.8)

B.S. Geographical Information Science (3.24/4.0) — Aug 2021 – Jun 2025

Core Courses: Principle of Geographical Information System (A+), Principle of Remote Sensing (A), GIS Application Development (A), Computer Programming (A), Quantitative Remote Sensing (A-), Data Structure (A-), Point Cloud Processing (A-)

Research Interests

Terrestrial ecosystem responses to climate extremes; carbon–water coupling; vegetation photosynthesis and productivity; hyperspectral remote sensing; solar-induced chlorophyll fluorescence; radiative transfer modeling; ecosystem modeling and model–data fusion.

Publications

Manuscripts under review

1. **Viewing Geometry Confounds the Satellite Detection of Vegetation Structural and Physiological Responses to Drought**
   Geophysical Research Letters
   Yan, W., Hao, D., Li, Q., Yan, X., Zhang, Z., Gao, Y., Li, K., Zhang, X., Hu, T., Huang, J., Matgen, P., & Zeng, Y.

Peer-reviewed papers

1. **Escape Ratio Contributes More Than Fluorescence Yield to the SIF–GPP Relationship Over Crops and Rainforest**
   IEEE Geoscience and Remote Sensing Letters (2024)
   Yan, W., Zeng, Y., Zhang, X., Zhao, W., Gao, Y., He, Y., & Hao, D.
2. **Tree-level carbon stock estimations across diverse species using multi-source remote sensing integration**
   Computers and Electronics in Agriculture (2025)
   Li, Q., Yin, J., Zhang, X., Hao, D., Ferreira, M. P., **Yan, W.**, Tian, Y., Zhang, D., Tan, S., Nie, S., An, T., Li, X., Huang, J., Su, W., & Zeng, Y.
3. **Mapping tree height in complex terrain of northern China using ultra-high-resolution images**
   Smart Agricultural Technology (2025)
   Li, Q., Zhang, X., Hao, D., **Yan, W.**, Zhao, Q., Tian, Y., & Zeng, Y.
4. **Global-PCG-10: a 10 m global map of plastic-covered greenhouses derived from Sentinel-2 in 2020**
   Earth System Science Data (2025)
   Niu, B., Feng, Q., Qiu, B., Su, S., Zhang, X., Cui, R., Zhang, X., Sun, F., **Yan, W.**, Zhao, S., Shi, H., Ou, C., Yan, X., Gong, J., Yin, G., Huang, J., Liu, J., Gao, B., Yao, X., Yang, J., & Zhu, D.

Research Experience

Ecosystem Remote Sensing — **Oct 2025 – May 2026**

Viewing Geometry Effects on Satellite Detection of Vegetation Responses to Drought — Supervisor: Prof. Yelu Zeng

- Investigated how satellite viewing geometry confounds the detection of vegetation structural and physiological drought responses across U.S. irrigated croplands, showing that observations depart from nadir and alter signal–GPP relationships.
- Integrated MODIS BRDF parameters, TROPOMI SIF, MODIS FPAR, ERA5-Land radiation, gridMET drought indices, and FluxSat/GEDI GPP products to reconstruct VIs and SIF metrics under observed, nadir, hotspot, and hemispherical geometries.

Ecosystem Remote Sensing — **Sep 2023 – May 2024**

Canopy Structural and Physiological Controls on the SIF−GPP Relationship — Supervisor: Prof. Yelu Zeng

- Investigated the relative contributions of canopy structural and physiological signals to the SIF−GPP relationship across crop and rainforest ecosystems and showed that the photon escape ratio explains LUE variability and the SIF−GPP relationship better than fluorescence quantum yield.
- Integrated TROPOMI SIF, MODIS vegetation indices, and FLUXCOM and CEDAR GPP products to derive LUE after quality control and cross-source harmonization.

Agricultural Plastic Mapping — **Nov 2022 – Apr 2024**

Global Mapping of Plastic-Covered Greenhouses Using Sentinel-2 — Supervisor: Prof. Quanlong Feng

- Contributed to a global-scale remote-sensing project that produced Global-PCG-10, the first global 10 m dataset of plastic-covered greenhouses, using multi-temporal Sentinel-2 imagery.
- Built samples and supported feature engineering, data preprocessing and large-scale implementation on Google Earth Engine for greenhouse detection and map generation.

Teaching Experience

- Teaching Assistant, Spatial Data Acquisition and Management (Undergraduate course) — Fall 2025
- Teaching Assistant, Quantitative Remote Sensing Technology and Applications (Graduate course) — Fall 2025

Conference Presentation

- Oral Presentation, Observation Geometry Effects on Detecting Vegetation Structural and Physiological Responses to Drought, 8th Asia-Oceania Group on Earth Observations (AOGEO) Workshop — 2026

Honors and Awards

- Merit-based Tuition Scholarship (¥8000), China Agricultural University — 2026
- Academic Excellence Scholarship (¥2000), China Agricultural University — 2026
- Esri China Cup – Remote Sensing Application Track (Excellence Prize) — 2023
- Esri China Cup – Map Storytelling Track (Third Prize) — 2023
- China Agricultural University Mathematical Modeling Competition (Third Prize) — 2023
- Renewable Energy Innovation Competition (Third Prize) — 2022

Skills & Interests

- Programming: Python, MATLAB, R, JavaScript, LaTeX, Markdown
- Remote Sensing: MODIS, Sentinel-2, Landsat, PACE, RTM
- Tools: Anaconda, Git, PyCharm, VSCode, Jupyter, GitHub, MATLAB, GEE
- Languages: Mandarin (native), English (IELTS)
- Instruments: ASD FieldSpec spectroradiometer, hemispherical canopy photography, SPAD 502, handheld infrared thermal sensor thermometer

References

- **Yelu Zeng**, Professor, China Agricultural University — zengyelu@cau.edu.cn
- **Dalei Hao**, Research Scientist, Pacific Northwest National Laboratory — dalei.hao@pnnl.gov
- **Quanlong Feng**, Associate Professor, China Agricultural University — fengql@cau.edu.cn